[COLLEGE / UNIVERSITY NAME PLACEHOLDER]

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY B.Tech Second Year | Course: Internet Programming Laboratory [Subject Code: _____]

ASSIGNMENT REPORT: CINEBOOK – MOVIE TICKET BOOKING WEBSITE

Academic Session: 2026–2027 **Faculty Guide:** [Faculty Name / Designation] **Project Type:** Full-Stack Group Web Application (5 Members) **Submission Date:** November 2026

1. Introduction & Objectives

CineBook is a full-stack, responsive movie ticket booking web application developed to satisfy the curriculum objectives of the B.Tech Internet Programming syllabus. The application bridges client-side interactive DOM interfaces with an asynchronous Node.js/Express REST backend and a relational MySQL database. The primary goals are: (a) enabling users to securely register, authenticate, and reserve cinema tickets with interactive seat selection; (b) validating transaction data and preventing double-booking; and (c) providing an administrative CRUD module to manage cinema catalogs.

2. Technologies Used

● **Frontend:** HTML5 Semantic Elements, CSS3 (Flexbox, Grid, Variables, @media print), Vanilla JavaScript (ES6+), DOM Manipulation, Fetch API. ● **Backend:** Node.js runtime, Express.js REST server, express-session (cookie session state), bcryptjs (password hashing), dotenv. ● **Database:** MySQL relational database, mysql2 promise pool, parameterized SQL queries preventing SQL Injection.

3. Explanation of the Five Connected Webpages

1. Page 1 – Registration (signup.html): Features full name, email, and password validation with regex checks, equality confirmation, and minimum length enforcement. Passwords are salted and hashed with bcrypt on the server before database insertion to prevent duplicate account creation. **2. Page 2 – Login (login.html):** Authenticates user credentials against stored MySQL password hashes. Upon verification, express-session assigns an HTTP-only session cookie (connect.sid) and redirects the user to the movie catalog or their pending booking. **3. Page 3 – Movie Browsing (movies.html):** Dynamically fetches movie items from GET /api/movies and constructs DOM cards displaying title, genre, duration, pricing, and posters. Includes real-time search and genre filtering without page reloads. **4. Page 4 – Seat Selection & Booking (booking.html):** Presents show date and time slot pickers alongside an interactive 5x8 cinema seat grid (Rows A–E, 40 seats). Occupied seats are queried from the backend and disabled. Calculates total price dynamically and revalidates prices on the server. **5. Page 5 – Ticket Confirmation & Printing (confirmation.html):** Fetches verified booking data via GET /api/booking/:id with strict user ownership protection. Displays a stylized cinema ticket with barcode simulation and triggers browser printing via JavaScript window.print() formatted with @media print CSS rules. **6. Module IV Bonus – Admin CRUD (admin.html):** Role-protected dashboard allowing administrators to Create, Read, Update, and Delete movies and monitor bookings.

4. MySQL Database Schema Explanation

The relational database "cinebook" is structured into three normalized tables with foreign key constraints: ● **users table:** user_id (INT, PK, AI), name (VARCHAR), email (VARCHAR, UNIQUE), password (VARCHAR, bcrypt hash), role (VARCHAR, default "user"). ● **movies table:** movie_id (INT, PK, AI), title (VARCHAR), description (TEXT), genre (VARCHAR), duration (INT min), price (DECIMAL), poster (VARCHAR). ● **bookings table:** booking_id (INT, PK, AI), user_id (FK to users), movie_id (FK to movies), show_date (DATE), show_time (VARCHAR), seats (VARCHAR), ticket_count (INT), total_price (DECIMAL), created_at (TIMESTAMP).

5. Individual Contributions of All Five Group Members

Member	Student Name / Roll	Assigned Page	Implemented Features & Syllabus Concepts
Member 1	[Student 1 Name] [Roll Number 1]	Page 1: Registration (signup.html)	Form validation (regex, length, equality), Fetch POST /api/signup, bcrypt hashing, error alert handling, redirect.
Member 2	[Student 2 Name] [Roll Number 2]	Page 2: Login (login.html)	Authentication via POST /api/login, express-session state, HTTP-only cookie, redirect logic, feedback banners.
Member 3	[Student 3 Name] [Roll Number 3]	Page 3: Movie Catalog (movies.html)	Dynamic card DOM creation from GET /api/movies, client search and genre filtering, responsive grid styling.
Member 4	[Student 4 Name] [Roll Number 4]	Page 4: Seat Booking (booking.html)	Interactive 5x8 seat map, booked seat detection, client-side dynamic pricing, login check guard, server revalidation.
Member 5	[Student 5 Name] [Roll Number 5]	Page 5: Ticket Output (confirmation.html)	Booking retrieval via GET /api/booking/:id, ticket layout, window.print() integration, @media print formatting.

6. Conclusion

The CineBook project successfully fulfills all curriculum requirements for the B.Tech Internet Programming laboratory coursework. By combining clean semantic HTML5, modern CSS3 styling, vanilla JavaScript DOM interactivity, Node.js/Express REST APIs, and a relational MySQL database, the project demonstrates real-world software engineering practices. All 16

automated backend test assertions passed, guaranteeing security, price integrity, and double-booking prevention. The codebase is modular, clean, and easily explainable in an academic viva.

Student Signatures: (1) _____ (2) _____ (3) _____ (4) _____ (5) _____ Faculty Examiner Signature: _____ Date: _____